

TeleAutoHINTS: A Virtual or Augmented Reality (VR/AR) System for Automated Tele-Neurologic Evaluation of Acute Vertigo

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1 INTRODUCTION

Dizziness/vertigo affects millions of patients in the U.S. annually [1], [2]. In acute settings, distinguishing rapidly and accurately between benign inner ear diseases and potentially debilitating neurologic injuries in patients experiencing a sudden onset of constant dizziness has garnered significant attention in neuro-otology and neuro-ophthalmology [5]. High-resolution brain imaging often fails to accurately differentiate these groups during the critical window for urgent interventions [6]. The inadequacies of brain imaging have contributed to increased healthcare spending on unnecessary tests and increased patient disability [2]. As a result, experts developed a three-step testing battery known as HINTS (Head Impulse, Nystagmus, and Test of Skew). This test is grounded in the physiology of the neural circuits connecting the inner ear's rotation and gravity sensors to the brain [4,7,8]. It evaluates changes in eye movement based on gaze positions, the vertical alignment of the eyes within the orbit, and subtle eye movement reactions to high velocity/low amplitude operator-induced head rotations. In the appropriate time frame for patients with AVS, HINTS has proven to be more sensitive and specific than a brain MRI in differentiating strokes from vestibular neuritis (inner ear disease) [6]. However, the widespread adoption of the HINTS test faces challenges. The primary challenge is the availability of vestibular neurology experts to interpret or administer the test and the scarcity of skilled emergency room providers trained in the technique [9,10,11]. This dearth of skilled individuals has prompted a reimagining of the HINTS assessment, leading to the popular use of portable video-oculography systems [12] and various telemedicine solutions [13,14,15,16]. Yet, none of these fully address the root problem. Consequently, we propose TeleAutoHINTS, a virtual/augmented reality solution for the automated tele-assessment of acute vertigo patients. We also conduct preliminary patient testing to evaluate the feasibility of this innovative approach. TeleAutoHINTS (Fig. 1) has the potential to reduce the need for skilled operators. The concept of remote sensing of head position and eye movement with AR and VR guidance has further clinical implications beyond assessing acute vertigo.

2 MATERIALS AND METHODS

The TeleAutoHINTS system comprises two parts: the head-mounted display-based testing platform (Patient Testing Interface [PTI]), designed to guide patients through the test, and the desktop visual-

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The Vision for TeleAutoHINTS

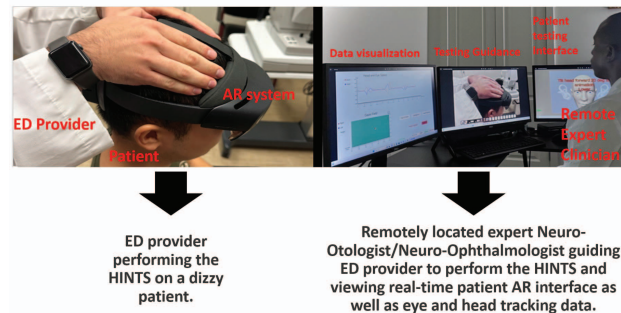


Figure 1: Concept of TeleAutoHINTS: A Virtual or Augmented Reality System for Automated Tele-Assessment of Acute Vertigo. ED provider - Emergency Room Medical Personnel (e.g., physician, physician assistant, nurse practitioner).

ization and validation platform (Medical Provider Interface [MPI]). The MPI allows providers to switch between automatic data collection (patient-initiated) and quasi-automatic data collection (provider-initiated). It also offers an interface for real-time monitoring and post-test data analysis. Both interfaces were created using Unity3D (version 2021.3.28f1).

The PTI was implemented on Microsoft HoloLens 2. We leveraged the HoloLens' built-in eye tracker to collect gaze data, and built-in gyroscope to collect angular velocity. We obtained the gaze data stream from each eye using the API provided by Microsoft at 60 fps.

The MPI is divided into two primary components: the User Interface and Visualization, and the Communication Portal. The User Interface and Visualization system is composed of three primary elements: Gaze Visualization (GV), Connection Control Panel (CCP), and Test Control Panel (TCP), as shown in the Video. The GV displays the pupil position data from both eyes for the test of skew (vertical only) and nystagmus tests (horizontal and vertical). For the head impulse test, it displays head and eye velocity-time graphs that show the head thrust and its corresponding opposite direction eye movement response.

3 USER STUDY

We conducted a preliminary study to determine if our TeleAutoHINTS system could collect meaningful data from patients.

3.1 Participants

All participants (n=3) were adults (≥ 18 years old) undergoing in-person video-oculography (VOG) and video head-impulse (vHIT) testing due to complaints related to various neurological illnesses. All participants consented to participate in the research. The study received approval from the institutional review board. All subjects

underwent their clinical VOG and vHIT tests before using TeleAutoHINTS.

3.2 TeleAutoHINTS Results

As shown in Fig. 2, the TeleAutoHINTS test results from all three subjects demonstrate that the system can produce high-quality head and eye traces that mimic physiologic eye movement responses to head impulses with the appropriate amplitude, velocity, and duration. The head velocity data from the head impulse exceeds 120 degrees per second for all subjects, aligning with clinical standards. This is crucial since low-velocity head impulses can yield inaccurate vestibular function information. The simulated nystagmus waveforms are robust, closely resembling the typical jerk nystagmus waveform morphology with an initiating slow phase followed by a fast phase in the opposite direction. This alternating pattern of slow and fast phases (or a beat) allows medical providers to assess: 1) the direction of the nystagmus (named for the direction of the fast phase), 2) the side of the brain or inner ear affected (indicated by the direction of the slow phase), and 3) the slow-phase velocity and amplitude of the nystagmus, which can aid in differentiating between brain and inner ear nystagmus in both fixation and fixation-removed scenarios. Lastly, the test of skew measures the degree of ocular misalignment in terms of vertical eye rotation.

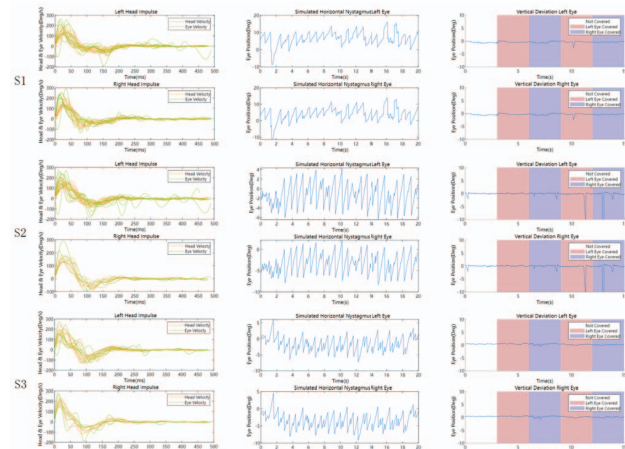


Figure 2: HINTS testing results for the three subjects. All subjects had normal head-impulse test defined as an eye to head velocity ratio of ≥ 0.7 . All 3 nystagmus traces demonstrate fast-phases to the right (upward deflection) or right-beating nystagmus. Subject 2 had the largest amplitude of all three. All three tests of skews were normal with artifacts (blinks) in all three mimicking vertical deviation of the eyes.

4 CONCLUSION

Traditional HINTS methodologies, while clinically invaluable, face significant challenges, particularly in emergency settings due to the lack of specialized knowledge and the absence of VOG tools. We have demonstrated that it is feasible to automatically collect components of the HINTS using our AR/VR-powered TeleAutoHINTS system. TeleAutoHINTS, harnessing the potential of AR/VR, addresses these challenges with its telemedicine-enabled self-directed protocols, modernizing essential HINTS components. As technology advances, the role of AI is set to expand, providing more streamlined analysis, especially in facilities grappling with expert shortages. By potentially integrating a wide range of diagnostic methods, TeleAutoHINTS stands to revolutionize acute vestibular syndrome evaluations

in emergency situations. Furthermore, this system is poised to become an invaluable tool for specialists like neuro-ophthalmologists, neuro-otologists, strabismus surgeons, and others involved in clinical roles or research within the realms of vestibular neurology and eye movement disorders.

REFERENCES

- [1] P. D. Sloane, "Dizziness in primary care. Results from the National Ambulatory Medical Care Survey," *J Fam Pract*, vol. 29, no. 1, Art. no. 1, Jul. 1989.
- [2] A. S. Saber Tehrani et al., "Rising Annual Costs of Dizziness Presentations to U.S. Emergency Departments," *Acad Emerg Med*, vol. 20, no. 7, Art. no. 7, Jul. 2013, doi: 10.1111/acem.12168.
- [3] C. S. K. Cheung et al., "Predictors of important neurological causes of dizziness among patients presenting to the emergency department," *Emerg Med J*, vol. 27, no. 7, Art. no. 7, Jul. 2010, doi: 10.1136/emj.2009.078014.
- [4] J. C. Kattah, A. V. Talkad, D. Z. Wang, Y.-H. Hsieh, and D. E. Newman-Toker, "HINTS to diagnose stroke in the acute vestibular syndrome: three-step bedside oculomotor examination more sensitive than early MRI diffusion-weighted imaging," *Stroke*, vol. 40, no. 11, Art. no. 11, Nov. 2009, doi: 10.1161/STROKEAHA.109.551234.
- [5] J. R. Hotson and R. W. Baloh, "Acute Vestibular Syndrome," *New England Journal of Medicine*, vol. 339, no. 10, pp. 680–685, Sep. 1998, doi: 10.1056/NEJM199809033391007.
- [6] D. E. Newman-Toker et al., "HINTS outperforms ABCD2 to screen for stroke in acute continuous vertigo and dizziness," *Acad Emerg Med*, vol. 20, no. 10, Art. no. 10, Oct. 2013, doi: 10.1111/acem.12223.
- [7] G. M. Halmagyi et al., "The human horizontal vestibulo-ocular reflex in response to high-acceleration stimulation before and after unilateral vestibular neurectomy," *Exp Brain Res*, vol. 81, no. 3, Art. no. 3, 1990, doi: 10.1007/BF02423496.
- [8] R. J. Leigh and D. S. Zee, *The neurology of eye movements*, 5th edition. In *Contemporary neurology series*, no. 90. Oxford ; New York: Oxford University Press, 2015.
- [9] C. Dmitriev et al., "Diagnostic Accuracy of the HINTS Exam in an Emergency Department: A Retrospective Chart Review," *Academic Emergency Medicine: Official Journal of the Society for Academic Emergency Medicine*, vol. 28, no. 4, pp. 387–393, Apr. 2021, doi: 10.1111/acem.14171.
- [10] M. Byworth, P. Johns, A. Pardhan, K. Srivastava, and M. Sharma, "Factors influencing HINTS exam usage by Canadian Emergency Medicine Physicians," *Can J Emerg Med*, vol. 24, no. 7, pp. 710–718, Nov. 2022, doi: 10.1007/s43678-022-00365-2.
- [11] R. Ohle et al., "Can Emergency Physicians Accurately Rule Out a Central Cause of Vertigo Using the HINTS Examination? A Systematic Review and Meta-analysis," *Academic Emergency Medicine: Official Journal of the Society for Academic Emergency Medicine*, vol. 27, no. 9, pp. 887–896, Sep. 2020, doi: 10.1111/acem.13960.
- [12] J. C. Kattah and D. E. Newman-Toker, "Video-Oculography to Guide Neuroimaging for Dizziness and Vertigo," *JAMA Otolaryngol Head Neck Surg*, vol. 148, no. 5, p. 474, May 2022, doi: 10.1001/jamaoto.2022.0330.
- [13] K. E. Green et al., "Opinion and Special Articles: Remote Evaluation of Acute Vertigo: Strategies and Technological Considerations," *Neurology*, vol. 96, no. 1, Art. no. 1, Jan. 2021, doi: 10.1212/WNL.0000000000010980.
- [14] P. Müller-Barna et al., "TeleVertigo: Diagnosing Stroke in Acute Dizziness: A Telemedicine-Supported Approach," *Stroke*, vol. 50, no. 11, Art. no. 11, Nov. 2019, doi: 10.1161/STROKEAHA.119.026505.
- [15] A. G. Shaikh et al., "Consensus on Virtual Management of Vestibular Disorders: Urgent Versus Expedited Care," *Cerebellum*, Aug. 2020, doi: 10.1007/s12311-020-01178-8.
- [16] R. J. So et al., "Quantitative vestibular assessment: The development and validation of a novel, remote video head impulse test against in-clinic measurements," *Laryngoscope Investig Oto*, vol. 8, no. 3, pp. 758–762, Jun. 2023, doi: 10.1002/lio2.1069.